Luis Carlos Rodríguez Quirós

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Professional Profile

Backend-focused Software Engineer with a Bachelor's Degree in Computer Engineering from Instituto Tecnológico de Costa Rica and hands-on experience developing RESTful APIs, backend functionality, and database solutions. Experienced with JavaScript, Python, Flask, PostgreSQL, SQL, Node.js, and Express.js, with practical experience designing and developing APIs, writing SQL queries, testing application functionality, and troubleshooting software issues. Experienced working with Git/GitHub, Postman, Jira, and Agile/Scrum environments, collaborating with development teams to build and maintain web applications. Strong interest in backend development, API development, and database engineering.

Education

Instituto Tecnológico de Costa Rica

Bachelor's Degree in Computer Engineering — *2016 - 2025*

Experience

Backend Developer - Internship — June 2024 - November 2024

KuarchTech Solutions S.A. — *San José, Costa Rica*

- Developed and maintained RESTful APIs using Python and Flask for a platform designed to support communication,group management, and organizational administration for companies and educational institutions.
- Developed backend functionality supporting hierarchical organizational structures, including administrators, teachers, students, groups, classes, assignments, attendance, grades, recorded class sessions, and group files.
- Designed and implemented 15+ PostgreSQL tables and relationships to support application requirements and improve data organization and scalability.
- Wrote SQL queries to create, retrieve, and manage application data while supporting backend functionality and database operations.
- Analyzed the existing database architecture and optimized the data model to improve scalability, maintainability, and reduce unnecessary data redundancy.
- Developed and tested API functionality using Postman to verify endpoints and application behavior.
- Refactored existing backend modules to support new functionality while maintaining compatibility with the existing system.
- Used Git and GitHub for version control and collaborated with other developers during the development and maintenance of backend functionality.
- Used Jira to track development tasks, coordinate work, and follow project requirements and progress.
- Participated in application testing and troubleshooting to identify functional issues and verify expected system behavior.

Technical Support Representative — March 2023 – July 2024

WStudio — *San José, Costa Rica*

- Provided technical support to approximately 20 customers per day, diagnosing and troubleshooting application issues affecting different areas of the platform.
- Investigated interface errors, application functionality issues, payment-related problems, account configuration issues, and user access problems.
- Reproduced and analyzed reported issues to identify possible causes and determine appropriate solutions or escalation paths.
- Followed up on reported technical issues to monitor their progress and communicate updates to customers until resolution or escalation.
- Collaborated with development teams by documenting and communicating recurring application issues and relevant technical details.
- Created and maintained issue reports using Redmine and Trello to track reported problems, follow-ups, and resolutions.
- Assisted customers with account configuration, access, payment-related issues, and general application functionality.

Full Stack Developer – Internship May 2023 - August 2023

FutPlay *San José, Costa Rica*

- Developed RESTful CRUD APIs using JavaScript to support application functionality and communication between the frontend and backend.
- Developed API endpoints for retrieving and managing customer information and court reservation data.
- Implemented backend functionality for managing customer records and court scheduling information using PostgreSQL.
- Wrote SQL queries to retrieve and manage application data stored in PostgreSQL.
- Tested API endpoints and functionality using Postman to verify request and response behavior.
- Reviewed existing code and API implementations to identify unnecessary components and opportunities for improvement.
- Analyzed backend and database interactions to improve data retrieval and API implementation.
- Participated in Scrum meetings to discuss project progress, coordinate development tasks, and communicate technical updates.

Projects

CRM System | *React, Node.js, Express, MongoDB, JavaScript*

- Designed and developed a full-stack CRM application for managing leads, clients, and sales pipelines.
- Developed RESTful APIs using Node.js and Express.js for user management, lead tracking, and status updates.
- Designed MongoDB schemas supporting dynamic pipeline stages and state transitions.
- Implemented authentication and role-based access control.
- Structured backend functionality using modular architecture principles.
- Developed a React frontend to interact with backend services and customer data.

Real-Time Video Communication System | *WebRTC + Node.js + Socket.IO*

- Built a real-time video communication platform using WebRTC, Socket.IO, and Node.js.
- Implemented signaling and peer connection management for multi-user communication.
- Configured STUN/TURN servers for NAT traversal and connection reliability.
- Managed room-based real-time communication logic.

Technical Skills

Programming Languages: JavaScript, Python, Java, SQL, HTML, CSS
Backend Development: Node.js, Express.js, Flask, RESTful APIs
Frontend Development: React
Databases: PostgreSQL, MongoDB, Oracle, SQL Server
Developer Tools: Git, GitHub, Postman, Docker, Jira, Linux
Technical Concepts: Database Design, Relational Data Modeling, CRUD Operations, Agile/Scrum

Languages

Spanish: Native.
English: Upper-intermediate (B2).